

Exam 3 Cheat Sheet

Speed of light in a vacuum:

$$c = 3.0 \times 10^8 \text{ m/s} \quad (\text{in a vacuum})$$

Speed of light where n is the index of refraction:

$$v = \frac{c}{n} \quad (\text{speed of light})$$

Law of refraction:

$$n_1 \sin \theta_1 = n_2 \sin \theta_2 \quad (\text{Snell's Law})$$

When light refracts along the boundary ($n_2 > n_1$, light traveling from less to more dense):

$$\theta_c = \sin^{-1}\left(\frac{n_2}{n_1}\right) \quad (\text{Critical angle})$$

Beam of light going through polarizing sheets:

$$I_1 = \frac{1}{2} I_0 \quad (\text{one half rule})$$

Going through another sheet:

$$I = I_1 \cos^2(\Delta\theta) \quad (\text{cosine-squared})$$

When a ray of light reflected by the film back to air undergo phase shift?

Yes, undergoing a phase shift of π ($\frac{1}{2}\lambda$) if light goes from a low index to high index. ($n_1 < n_2$)

No if it goes from high to low index. ($n_2 > n_3$)

Minima for Destructive interference:

$$2L = m \frac{\lambda}{n_2} \quad (1)$$

Minima for Constructive interference:

$$2L = \left(m + \frac{1}{2}\right) \frac{\lambda}{n_2} \quad (2)$$

If in Phase: Minima for Destructive interference:

$$2L = \left(m + \frac{1}{2}\right) \frac{\lambda}{n_2} \quad (3)$$

Minima for Constructive interference:

$$2L = m \frac{\lambda}{n_2} \quad (4)$$

Double slit: $\sin \theta$ is so small that:

$$\sin \theta = \tan \theta = \frac{y_m}{D} \quad (5)$$

interference Minima:

$$d \sin \theta = \left(m + \frac{1}{2}\right) \lambda \quad (6)$$

interference Maxima:

$$d \sin \theta = m \lambda \quad (7)$$

Path Difference (use $m + \frac{1}{2}$ if looking for):

$$\Delta L = d \sin \theta = m \lambda \quad (8)$$

Phase Difference:

$$\Delta\theta = \frac{2\pi}{\lambda} \Delta L \quad (9)$$

Spacing between adjacent bright fringes:

$$y_{m1} - y_{m0} = \lambda \frac{D}{d} \quad (10)$$

Single Slit: Diffraction Minima

$$a \sin \theta = m\lambda \quad (11)$$

Find width of central bright fringe:

Remember that $\tan \theta = \text{oppo} / \text{adj}$, we can use where $2y_1$ is the width:

$$2y_1 = 2L \tan \theta \quad (12)$$

Rayleigh's Criterion:

$$\theta_R = 1.22\left(\frac{\lambda}{d}\right) \quad (13)$$

Length between eye and object:

$$L = \frac{Dd}{1.22\lambda} \quad (14)$$

Where D is size of object, and d is the diameter of eye.

Grating Size:

$$d = \frac{W}{N} \quad (15)$$

Diffraction Maxima:

$$d \sin \theta = m\lambda; \quad (16)$$

Diffraction Minima:

$$d \sin \theta = \left(m + \frac{1}{2}\right)\lambda; \quad (17)$$

Mirrors and Images:

Focal length:

$$f = \frac{1}{2}r \quad (18)$$

Object distance P, image distance i, focal length f:

$$\frac{1}{f} = \frac{1}{p} + \frac{1}{i} \quad (19)$$

Magnification:

$$m = \frac{h'}{h} = \frac{-i}{p} \quad (20)$$

For images: